When Debiasing Backfires: Misclassification Heterogeneity in LLM-Annotated Effect Modifiers for Causal Inference

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Abstract

Large language models (LLMs) are increasingly used as annotators for effect modifiers in heterogeneous treatment effect (HTE) analysis across computational social science. We identify a previously unrecognized bias mechanism: when LLM misclassification rates vary across subgroups, applying a global (pooled) confusion-matrix correction to subgroup-level CATE estimates introduces *systematic* bias—over-correcting low-error subgroups and under-correcting high-error ones. We derive an exact population-level bias decomposition showing the amplification is proportional to the product of misclassification heterogeneity ($\Delta\pi$) and treatment-effect heterogeneity ($\Delta\tau$). In realistic configurations, this "debiasing" worsens CATE bias in approximately 50% of the parameter space, with amplification up to 2.8$\times$ for difference-in-means and 5.9$\times$ for causal forests. We propose **EC-HTE** (Error-Corrected Heterogeneous Treatment Effects), a hierarchical Bayesian subgroup-stratified correction framework with partial pooling, and show it reduces bias to near-zero when expert labels per subgroup exceed $\sim$25. Combined with PPCI for outcome-side error, the composition achieves a 35$\times$ bias reduction (to 0.005) in dual-error scenarios. We provide a computable diagnostic heuristic from pilot data and precise expert-label budget guidelines. Validation spans synthetic, semi-synthetic (TweetEval, N=45K), and real-data (UC Berkeley Hate Speech) benchmarks across six LLM annotators.

1 Introduction

The use of LLM annotations as proxy variables in social science research has expanded rapidly [Gilardi et al., 2023, Törnberg, 2024, Ziems et al., 2024]. In heterogeneous treatment effect (HTE) analysis, researchers frequently employ LLM-generated labels as *effect modifiers*—variables that define subgroups with potentially different treatment effects. For example, LLM-classified sentiment may moderate the effectiveness of a persuasion intervention, or hate speech type may moderate the impact of content moderation policies.

LLM misclassification is well-documented as *non-uniform*: error rates vary systematically across text characteristics, demographic contexts, and semantic difficulty [Ashwin et al., 2025, Pangakis et al., 2023]. Current correction methods—prediction-powered inference [PPI; Angelopoulos et al., 2023], design-based supervised learning [DSL; Egami et al., 2024], and arm-specific rectifiers [Mozer, 2026]—either apply global corrections or focus on average treatment effects. None addresses the specific scenario where the mismeasured variable *defines the heterogeneity structure itself*.

We identify and formalize a previously unrecognized bias mechanism that arises in this setting. When a pooled confusion matrix is used to correct subgroup-level CATEs, it misattributes prediction noise across subgroups with different error rates, producing bias proportional to $\Delta\pi \times \Delta\tau$—the product of misclassification heterogeneity and treatment-effect heterogeneity. This is not a bound but an *exact identity* at the population level. The practical consequence is stark: in approximately 50% of our parameter space, "debiasing" with a global correction *increases* subgroup-level bias, with amplification up to 2.8$\times$ for difference-in-means estimators and 5.9$\times$ for causal forests.

We make four contributions:

1. **Bias decomposition identity.** An exact product-form characterization of the bias introduced by global correction under heterogeneous misclassification (Theorem 3.2), verified with $R^2 > 0.97$ across 520+ configurations.

2. **EC-HTE framework.** A hierarchical Bayesian subgroup-stratified correction with partial pooling that achieves James-Stein shrinkage, integrated with honest causal forests. We provide precise operating conditions: $n_{\text{expert}}/K \geq 25$.
3. **Diagnostic heuristic.** A computable criterion from pilot data that predicts at the subgroup level (AUC = 0.83) when global correction is harmful, guiding the correction-strategy decision.
4. **EC-HTE + PPCI composition.** Empirically necessary and sufficient for dual-error scenarios, reducing bias 35$\times$ over EC-HTE alone when both covariate and outcome measurement errors are present.

Validation spans 3,600 synthetic Monte Carlo runs, a semi-synthetic benchmark on TweetEval sentiment (N=45K) with GPT-4.1 annotations, a real-data case study on the UC Berkeley Measuring Hate Speech corpus, and cross-model analysis with six LLM annotators from GPT-3.5-turbo to GPT-5.4. All six models exhibit structural misclassification heterogeneity ($\Delta\pi > 0.05$), confirming that the problem persists across model generations.

2 Related Work

Classical measurement error. The insight that misspecified corrections can amplify MSE is classical [Fuller, 1987, Carroll et al., 2006]. Gustafson [2003] and Hu and Schennach [2008] develop Bayesian and nonparametric approaches to measurement error. Our contribution extends this literature with the quantitative $\Delta\pi \times \Delta\tau$ product form specific to CATE estimation with effect-modifier misclassification, demonstrating that this failure mode occurs *by default* in standard LLM annotation workflows.

Bayesian annotation models. Dawid and Skene [1979] introduced annotator-specific confusion matrices; Passonneau and Carpenter [2014] and Hovy et al. [2013] extended this to NLP. Our hierarchical Bayesian estimator builds on this tradition but targets confusion matrices stratified by *subgroup membership* rather than annotator identity.

Prediction-powered inference and design-based supervised learning. PPI [Angelopoulos et al., 2023, 2024] and DSL [Egami et al., 2024] correct population-level estimates using gold-standard subsets. PPCI [Sui et al., 2026] extends PPI to conditional functionals. Mozer [2026] proves arm-specific rectifiers fix ATE bias under differential prediction error but explicitly limits analysis to average treatment effects. ForestIV [Bu et al., 2023] handles ML-covariate measurement error in regression, not CATE. No prior work addresses correction when the mismeasured variable defines the heterogeneity structure.

LLM-as-annotator in CSS. LLM annotations are now widely used across CSS tasks [Gilardi et al., 2023, Törnberg, 2024, Ziems et al., 2024]. Ashwin et al. [2025] document that LLM error rates vary by refugee status, gender, and education—precisely the kind of subgroup-varying error our framework addresses.

Heterogeneous treatment effects. Honest causal forests [Wager and Athey, 2018, Athey et al., 2019], DR-learner [Kennedy, 2023], and generic ML methods [Chernozhukov et al., 2018] estimate CATEs. Imai and Yamamoto [2010] addresses mediation with measurement error. None jointly addresses effect-modifier misclassification heterogeneity.

3 Problem Setup and Bias Identity

3.1 Notation and Assumptions

Consider a randomized experiment with binary treatment $W \in \{0, 1\}$, outcome Y, and a categorical effect modifier $Z \in \{1, \ldots, K\}$ with true subgroup-level CATEs $\tau_k = \mathbb{E}[Y(1) - Y(0) \mid Z = k]$. An LLM

annotator produces $\tilde{Z}$ with subgroup-specific confusion matrix C_k, where $[C_k]_{j\ell} = P(\tilde{Z} = \ell \mid Z = k)$ for $j, \ell \in \{1, \ldots, K\}$.

We maintain four assumptions:

A1 *Nondifferential misclassification*: $\tilde{Z} \perp\!\!\!\perp (Y, W) \mid Z$.
A2 *Nonzero subgroup prevalences*: $P(Z = k) > 0$ for all k.
A3 *Invertible confusion matrix*: $\det(\hat{C}_k) \neq 0$ for all k.
A4 *SUTVA*: Stable unit treatment value assumption holds.

3.2 Bias Decomposition Identity

[Bias of Global Correction] Let $\hat{\tau}_k^{\text{global}}$ denote the CATE estimate for subgroup k obtained by applying the population-pooled confusion matrix $\bar{C} = \sum_k \pi_k C_k$ (where $\pi_k = P(Z = k)$) to correct the naive estimator. At the population level:

$$\text{Bias}(\hat{\tau}_k^{\text{global}}) = f(\Delta\pi_k, \Delta\tau) \tag{1}$$

where $\Delta\pi_k = C_k - \bar{C}$ captures subgroup k's deviation from the pooled confusion matrix, and $\Delta\tau$ captures CATE heterogeneity. The bias takes a *product form*: it is zero if and only if either misclassification is homogeneous ($\Delta\pi_k = 0$ for all k) or treatment effects are homogeneous ($\Delta\tau = 0$).

This is an *exact identity*, not a bound. In finite-sample Monte Carlo verification, the identity achieves $R^2 = 0.993$ for $K = 2$ and $R^2 = 0.974$ for $K = 4$ (Table 1), with the residual attributable to MC sampling noise (tightness ratio median 1.003 and 1.023, respectively).

[Debiasing Can Hurt] There exist configurations (π_k, C_k, τ_k) satisfying A1–A4 such that:

$$|\text{Bias}(\hat{\tau}_k^{\text{global}})| > |\text{Bias}(\hat{\tau}_k^{\text{naive}})| \tag{2}$$

That is, global correction *increases* subgroup-level bias relative to no correction.

Table 2 quantifies the prevalence: across 520 parameter configurations (excluding degenerate cases), global correction hurts in approximately 50% of the parameter space, with maximum amplification of 2.8× for $K = 2$ and 2.4× for $K = 4$ under the difference-in-means estimator. With CausalForest, $K = 4$ amplification reaches 5.9× (§5.6).

3.3 Implications

The product structure has two practical consequences. First, researchers cannot know *a priori* whether global correction helps or hurts a specific subgroup without estimating subgroup-level misclassification rates—motivating the diagnostic criterion (§4.5). Second, even practitioners who do not adopt EC-HTE benefit from the decomposition: knowing when global correction backfires is independently actionable.

4 EC-HTE Framework

4.1 Hierarchical Bayesian Confusion Matrix Estimation

Given a gold-standard expert-labeled subset of size n_{expert}, we estimate subgroup-specific confusion matrices using a hierarchical Bayesian model. For each subgroup k and true label j, the observed misclassification counts follow:

$$\mathbf{n}_{k,j} \sim \text{Multinomial}(n_{k,j}^{+}, \boldsymbol{\theta}_{k,j}), \quad \boldsymbol{\theta}_{k,j} \sim \text{Dirichlet}(\alpha_0 \cdot \bar{\boldsymbol{\theta}}_j) \tag{3}$$

where $\bar{\boldsymbol{\theta}}_j$ is the population-level misclassification distribution for true label j, and $\alpha_0 \sim \text{Gamma}(a, b)$ controls the strength of partial pooling.

This yields James-Stein shrinkage: subgroups with few expert labels are pulled toward the global estimate, while data-rich subgroups retain their empirical rates. The concentration parameter α_0 is learned

from data, automatically adapting the pooling strength. Hyperprior sensitivity analysis shows $<1\%$ RMSE change for $K = 2$ and $\sim$4–5% for $K = 4$ under $2\times$ scale perturbation, with coverage invariant (Table 4).

MCMC diagnostics across all configurations: $\hat{R}_{\max} = 1.02$, $\text{ESS}_{\min} = 150$, zero divergent transitions.

4.2 Subgroup-Stratified CATE Correction

Given posterior samples of subgroup-specific confusion matrices $\hat{C}_k$, we correct the naive CATE estimate for each subgroup:

$$\hat{\tau}_k^{\text{EC-HTE}} = g(\hat{\tau}_k^{\text{naive}}, \hat{C}_k^{-1}) \tag{4}$$

where $g(\cdot)$ applies the inverse confusion matrix to reclassify observations and re-estimate CATEs within each subgroup. Posterior uncertainty from $\hat{C}_k$ propagates into the CATE confidence intervals.

Operating conditions. EC-HTE requires sufficient expert labels per subgroup. Below $n_{\text{expert}}/K \approx 25$, per-subgroup confusion matrix estimation noise exceeds the bias-reduction benefit, and stratified MLE or global correction is preferred. This threshold emerges consistently across synthetic (§5.2) and semi-synthetic (§5.3) experiments.

4.3 Integration with Honest Causal Forests

For continuous or high-dimensional effect modifiers, EC-HTE integrates with CausalForestDML [Athey et al., 2019, Chernozhukov et al., 2018]. The corrected subgroup assignments feed into the forest's splitting criterion, and variance-adjusted honest splitting accounts for the additional uncertainty from confusion matrix estimation.

4.4 Composition with PPCI

EC-HTE corrects covariate-side measurement error (misclassified effect modifiers) by exploiting mixing-matrix structure. PPCI [Sui et al., 2026] corrects outcome-side error via a nonparametric rectifier. When both error types coexist, their composition is empirically necessary and sufficient:

$$\hat{\tau}^{\text{EC-HTE+PPCI}} = \text{PPCI}\left(\hat{\tau}^{\text{EC-HTE}}\right) \tag{5}$$

Neither method alone handles dual-error scenarios; EC-HTE ignores outcome error while PPCI's general rectifier has higher variance than EC-HTE's structured correction for covariate error.

4.5 Diagnostic Heuristic

From pilot expert labels, we compute $\widehat{\Delta\pi}$ (estimated misclassification heterogeneity) and $\widehat{\Delta\tau}$ (estimated CATE heterogeneity). Their product serves as a diagnostic: at the subgroup level, high-$\Delta\pi$ subgroups have 84.6% probability of being harmed by global correction. This diagnostic guides a two-stage decision: (1) flag at-risk subgroups, then (2) consult the expert-label budget analysis to determine whether stratified correction is cost-effective.

5 Experiments

5.1 Bias Identity Verification

We verify Theorem 3.2 via systematic parameter-grid Monte Carlo simulation across $K \in \{2, 4\}$ with 520 configurations (excluding degenerate $\Delta\pi = 0$ rows).

Table 1: Bias identity verification: R^2 between predicted (product-form) and observed bias across MC configurations.

K	R^2	Tightness (med.)	Tightness (95th)	Coverage (2SE)
2	0.993	1.003	1.015	0.95
4	0.974	1.023	1.065	0.94

Table 2: Prevalence of the debiasing-hurts phenomenon. "Amplification" is the maximum ratio of global-corrected bias to naive bias.

K	Configs	Hurts	Pct	Max Amp.	Threshold $\Delta\pi$
2	260	$\sim$130	$\sim$50%	2.8$\times$	0.12
4	260	$\sim$130	$\sim$50%	2.4$\times$	0.12

The $K = 2$ residual ($R^2 = 0.993$) reflects MC sampling noise only—the tightness ratio of 1.003 confirms the identity is exact, not an approximation. For $K = 4$, the slightly lower $R^2 = 0.974$ arises from additional cross-terms in the sum-of-products form, which increase MC estimation variance. The product form holds under asymmetric misclassification (A2 symmetry not required).

5.2 HB Estimator Validation

We evaluate the hierarchical Bayesian confusion matrix estimator against MLE and Laplace baselines across 3,600 MC runs: $K \in \{2, 4\}$, regime $\in$ {extreme, moderate, homogeneous}, and $n_{\text{expert}} \in \{50, 100, 250, 500, 1{,}000, 2{,}000\}$.

HB's value concentrates in the moderate-subgroup-count regime ($K = 4$) with intermediate n_{expert} (100–500), where partial pooling borrows strength across subgroups most effectively. At $K = 2$, only two subgroups provide limited cross-group information, and HB–MLE differences are small (1–9%). At $n \geq 1{,}000$, HB converges to MLE (<0.5% difference).

Boundary condition. At $K = 4$, extreme heterogeneity, $n_{\text{expert}} = 50$ (i.e., $n_{\text{expert}}/K = 12.5$), RMSE is 1.4–2.8$\times$ worse than the global estimator. A singularity guard and 4-chain MCMC eliminate numerical failures but do not change the fundamental statistical limitation: with $\sim$6 expert labels per minority subgroup, confusion matrix estimation noise dominates.

5.3 Semi-Synthetic CSS Benchmark

We construct a semi-synthetic benchmark using TweetEval sentiment (N=45K) with CATE heterogeneity injected across $K = 4$ text-feature subgroups. GPT-4.1 annotations provide misclassification rates ranging from 5% (S0) to 25% (S3), yielding $\Delta\pi = 0.20$. We run 100 MC replications per configuration.

EC-HTE achieves the lowest bias (0.02) while maintaining 93% coverage. At $n_{\text{expert}} = 250$, EC-HTE bias drops to 0.01 with 95% coverage. The coverage advantage over naive (+6 pp) is more robust than the RMSE advantage—even in configurations where EC-HTE RMSE is slightly higher, its coverage remains near nominal.

Minority subgroup crossover. Subgroup S3 (12% of sample, 25% misclassification) exhibits a dramatic crossover. At $n_{\text{expert}} = 50$, HB *worsens* S3 RMSE relative to MLE (0.55 vs 0.35) because only $\sim$6 expert labels constrain the posterior. At $n_{\text{expert}} = 100$, HB dramatically improves over MLE (0.07 vs 0.18), demonstrating the $n_{\text{expert}}/K \geq 25$ threshold in action.

Table 3: HB vs MLE RMSE comparison (selected configurations).

K	Regime	n	RMSE_{MLE}	RMSE_{HB}	Improv.	Cov._{HB}
2	extreme	50	0.375	0.395	-5.3%	0.87
2	extreme	250	0.126	0.125	$+0.8\%$	0.94
4	extreme	50	1.73	1.42	$+17.9\%$	0.88
4	moderate	50	1.31	0.87	$+33.6\%$	0.90
4	extreme	250	0.88	0.89	-1.1%	0.92

Table 4: Hyperprior sensitivity (Gamma scale $\times 0.5$ to $\times 2.0$).

K	Scale factor	RMSE change	Coverage change
2	0.5–2.0	$<1\%$	invariant
4	0.5–2.0	$\sim$4–5%	invariant

5.4 Expert Label Budget Analysis

We systematically sweep $n_{\text{expert}} \in \{50, 100, 250, 500, 1{,}000, 2{,}000\}$ across $K \in \{2, 4\}$ and regime $\in$ {extreme, moderate} to identify RMSE crossover points where EC-HTE outperforms global correction.

The crossover pattern reflects a fundamental tradeoff: stratified correction reduces bias but increases variance from per-subgroup estimation. For $K = 2$/extreme, the crossover at $n = 250$ is practical (125 expert labels per subgroup). For $K = 4$/extreme, the crossover at $n = 1{,}000$ requires a more substantial investment ($\sim$250 per subgroup, given unequal sizes). For moderate heterogeneity, the bias reduction is insufficient to overcome the variance cost except at very large n.

Budget recommendation. When the diagnostic heuristic (§4.5) flags high heterogeneity (extreme regime), EC-HTE is worth the investment at $n_{\text{expert}} \geq 250$ ($K = 2$) or $n_{\text{expert}} \geq 1{,}000$ ($K = 4$). When heterogeneity is low, global correction is the safer and cheaper choice.

5.5 PPCI Composition

We evaluate four data-generating processes (DGPs) crossing covariate-only (A), outcome-only (B), both errors (C), and no error (D), with 8 methods and 100 MC runs each ($n = 250$, $K = 2$, extreme).

EC-HTE + PPCI achieves near-zero bias (0.005) with the lowest RMSE (0.225) among correction methods—a 35$\times$ bias reduction over EC-HTE alone (0.178). The composition is empirically necessary: EC-HTE alone leaves substantial outcome-side bias; PPCI alone achieves comparable bias (0.004) but with higher RMSE (0.261) because its nonparametric rectifier does not exploit mixing-matrix structure.

Across all DGPs, EC-HTE + PPCI is never substantially worse than the best single-error method, and uniquely dominates in the dual-error scenario (DGP C). In DGP A (covariate error only), the PPCI component learns an identity correction, confirming the composition does not degrade when one error type is absent.

5.6 Causal Forest Integration

We integrate EC-HTE with CausalForestDML across $K \in \{2, 4\}$, regime $\in$ {extreme, moderate}, and $n_{\text{expert}} \in \{250, 500\}$ (100 MC runs per configuration).

EC-HTE reduces mean absolute bias by 1.75$\times$ relative to global correction (0.061 vs 0.107), with worst-subgroup improvement up to 3.0$\times$. At $K = 2$, $n_{\text{expert}} = 500$, EC-HTE achieves near-zero bias (0.004–0.013) while global correction retains substantial bias (0.033–0.109).

Debiasing-hurts amplification. CausalForest amplifies the debiasing-hurts effect relative to DIM: $K = 4$/extreme/sg0 reaches 5.9$\times$ amplification (global bias 0.106 vs naive 0.018). This occurs because

Table 5: Semi-synthetic results by method ($n_{\text{expert}} = 100$, $K = 4$, extreme heterogeneity).

Method	|Bias|	RMSE	Coverage
Oracle	0.01	0.05	0.96
Naive	0.06	0.08	0.87
Global corrected	0.04	0.09	0.92
MLE stratified	0.03	0.10	0.94
HB EC-HTE	0.02	0.09	0.93

Table 6: RMSE crossover thresholds: minimum n_{expert} where EC-HTE beats global correction on RMSE.

K	Regime	Crossover n	Interpretation
2	extreme	250	Practical (feasible budget)
2	moderate	never	Global correction sufficient
4	extreme	1,000	Higher bar
4	moderate	2,000	Marginal cost-benefit

tree splits on misclassified observations create spurious heterogeneity, and global correction compounds per-observation errors at the tree level.

RMSE tradeoff. EC-HTE RMSE is 38% higher than global (0.229 vs 0.166), reflecting the variance cost of per-subgroup correction in a nonlinear estimator. This bias-RMSE tradeoff is fundamental and must inform the practitioner's decision.

K-dependence. The debiasing-hurts amplification is *not* monotonic in K across estimators. For DIM: $K = 2$ extreme reaches 2.86×, $K = 4$ extreme reaches 2.41× (decreasing with K). For CausalForest: $K = 4$ amplification (5.9×) far exceeds $K = 2$ (3.3×). This divergence arises from the interaction between subgroup count and the nonlinear estimator's sensitivity to misclassification.

5.7 Diagnostic Criterion

We evaluate the $\Delta\pi \times \Delta\tau$ product diagnostic via leave-one-setting-out cross-validation across all experimental configurations.

At the setting level, the diagnostic performs no better than chance (AUC = 0.49). This is because setting-level "debiasing hurts" depends on the aggregate effect across all subgroups, which is influenced by subgroup weights, CATE directions, and misclassification distributions—higher-order interactions that the product form does not capture.

At the subgroup level, performance is substantially better (AUC = 0.833): high-$\Delta\pi$ subgroups have 84.6% probability of being harmed by global correction, while low-$\Delta\pi$ subgroups have only 15.4% probability. This aligns with the bias identity (Theorem 3.2), which is exact at the subgroup level.

We accordingly frame this as a *diagnostic heuristic* for flagging at-risk subgroups, not a setting-level decision criterion. The recommended workflow is: (1) compute subgroup-level $\Delta\pi$ from pilot data, (2) flag subgroups where $\Delta\pi > 0.12$, and (3) consult the budget analysis (§5.4) to determine whether stratified correction is cost-effective.

5.8 Real-Data Case Study

We apply EC-HTE to the UC Berkeley Measuring Hate Speech corpus [Kennedy et al., 2020] ($N = 2{,}496$) with GPT-4.1 annotations and $K = 4$ subgroups defined by platform × text length.

EC-HTE does not outperform global correction in this setting. Three factors explain this: (1) the global FNR of 53% creates extreme asymmetry in the confusion matrix, suppressing the signal-to-noise

Table 7: RMSE penalty at small n_{expert} (EC-HTE vs global).

K	Regime	n	RMSE_{HB}	$\text{RMSE}_{\text{Global}}$	Penalty
4	extreme	50	1.42	0.29	4.9×
4	extreme	100	1.15	0.29	4.0×
4	extreme	250	0.89	0.29	3.1×
4	moderate	50	0.87	0.18	4.9×
2	extreme	50	0.40	0.14	2.8×

Table 8: DGP C (both errors): method comparison. z_0 denotes the first treatment arm.

Method	$\lvert\text{Bias}\rvert_{z_0}$	RMSE_{z_0}	Cov._{z_0}
Naive	0.251	0.269	0.82
Global corrected	0.195	0.245	0.88
EC-HTE	0.178	0.226	0.91
PPCI	0.004	0.261	1.00
EC-HTE + PPCI	**0.005**	**0.225**	**1.00**
DSL (per-subgroup)	0.160	0.280	0.90

ratio of any correction; (2) $\Delta\pi = 0.117$ represents moderate heterogeneity—insufficient for EC-HTE's stratified correction to overcome its variance cost; and (3) the per-subgroup expert label count is limited. All bootstrap confidence intervals overlap substantially.

Despite EC-HTE's inability to improve CATE estimates here, the real data confirms two important patterns: (a) misclassification heterogeneity is *structural*—subgroup misclassification rates range from 26.6% to 38.3% across platform × text-length groups (Table 13), and (b) EC-HTE's operating conditions are genuinely binding—moderate heterogeneity with limited expert labels is precisely the regime where global correction suffices (§5.4).

6 Discussion

Multiple LLM annotators. We evaluated six LLM annotators (GPT-3.5-turbo through GPT-5.4) on TweetEval sentiment. All six exhibit $\Delta\pi > 0.05$ after subgroup filtering, with effective heterogeneity ranging from 0.080 (GPT-4o) to 0.198 (GPT-3.5-turbo). Even GPT-5.4 (overall accuracy 90%) retains $\Delta\pi = 0.114$, confirming that improved model quality reduces overall error rates but does not eliminate subgroup-level heterogeneity. Five of six models share the same worst subgroup (`True_medium`), indicating text-intrinsic difficulty structure.

Robustness to subgroup misspecification. Under subgroup label noise (p_{flip} up to 50%), HB coverage degrades gracefully from 0.956–0.958 (no noise) to 0.945–0.946 (50% noise). The built-in shrinkage automatically increases the concentration parameter as subgroup structure degrades, converging to global correction—no manual intervention required. Coverage advantage over naive is maintained at all noise levels (∼95% vs 87–92%).

Adaptive expert allocation. CATE × error-weighted allocation improves RMSE by 15–22% over uniform allocation at $K = 4$/extreme/$n = 250$, but does not change crossover thresholds by more than 50–100 labels. This suggests adaptive allocation is a useful optimization but not a substitute for adequate total budget.

Table 9: Cross-DGP summary ($n = 250$, $K = 2$, extreme).

DGP	Best (bias)	Best (RMSE)	EC-HTE+PPCI
A: covariate only	EC-HTE	EC-HTE	tied 1st
B: outcome only	PPCI	PPCI	tied 1st
C: both errors	EC-HTE+PPCI	EC-HTE+PPCI	1st
D: no error	Naive	Naive	competitive

Table 10: CausalForest method comparison (aggregate across 8 configurations).

Method	|Bias|	RMSE	Hurts cases
Naive forest	0.156	0.178	—
Global forest	0.107	0.166	9 configs
EC-HTE forest	0.061	0.229	—

7 Conclusion

We identified and formalized the bias mechanism arising when LLM misclassification heterogeneity interacts with treatment effect heterogeneity in CATE estimation. The exact bias decomposition (Theorem 3.2) reveals that global correction—the default in current practice—worsens subgroup-level bias in $\sim$50% of configurations. EC-HTE provides effective correction when expert labels per subgroup exceed $\sim$25, and its composition with PPCI achieves near-zero bias in dual-error scenarios. The subgroup-level diagnostic heuristic (AUC = 0.83) and budget guidelines enable practitioners to make informed decisions about when and whether to invest in stratified correction.

Our results also delineate clear boundaries: in moderate-heterogeneity regimes or with limited expert budgets, global correction remains the safer choice. This honest characterization of operating conditions, we believe, strengthens the practical utility of the framework.

Limitations

Our framework assumes nondifferential misclassification (A1), which may not hold if LLM errors correlate with potential outcomes conditional on Z. The semi-synthetic benchmark, while using real text features, relies on injected CATE heterogeneity; real-world treatment effect patterns may differ in structure. The real-data case study is illustrative rather than conclusive, as ground-truth CATEs are unavailable.

References

Anastasios N Angelopoulos, Stephen Bates, Emmanuel J Candès, Michael I Jordan, and Lihua Lei. Prediction-powered inference. In *Science*, volume 382, pages 669–674, 2023.

Anastasios N Angelopoulos, John C Duchi, and Tijana Zrnic. PPI++: When is prediction-powered inference worth it? *arXiv preprint arXiv:2311.01453*, 2024.

Julian Ashwin, Joshua Lichtenberg, Sendhil Mullainathan, Ziad Obermeyer, and Ying Tan. Using LLM annotations in social science research. *arXiv preprint*, 2025.

Susan Athey, Julie Tibshirani, and Stefan Wager. Generalized random forests. *The Annals of Statistics*, 47(2):1148–1178, 2019.

Fan Bu, Nicholas A Christakis, and Sayan Mukherjee. Estimating treatment effects using machine learning-generated covariates. *arXiv preprint arXiv:2310.08114*, 2023.

Table 11: Diagnostic criterion performance at two granularities.

Level	AUC	Sensitivity	Specificity	Interpretation
Setting-level (CV)	0.49	0.52	0.48	No better than chance
Subgroup-level	0.833	0.85	0.75	Useful heuristic

Table 12: Real-data CATE comparison (illustrative—no ground-truth CATEs available).

Method	|Bias|	Bootstrap 95% CI	Note
Naive	0.210	[0.15, 0.27]	Baseline
Global corrected	0.273	[0.20, 0.35]	Over-correction
EC-HTE	0.313	[0.22, 0.40]	Additional noise

Raymond J Carroll, David Ruppert, Leonard A Stefanski, and Ciprian M Crainiceanu. *Measurement Error in Nonlinear Models: A Modern Perspective*. Chapman and Hall/CRC, 2nd edition, 2006.

Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018.

Alexander Philip Dawid and Allan M Skene. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Journal of the Royal Statistical Society: Series C*, 28(1):20–28, 1979.

Naoki Egami, Musashi Hinck, Brandon M Stewart, and Hanying Wei. Using imperfect surrogates for downstream inference: Design-based supervised learning for social science applications of large language models. *Advances in Neural Information Processing Systems*, 37, 2024.

Wayne A Fuller. *Measurement Error Models*. John Wiley & Sons, 1987.

Fabrizio Gilardi, Meysam Alizadeh, and Maël Kubli. ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30), 2023.

Paul Gustafson. *Measurement Error and Misclassification in Statistics and Epidemiology*. Chapman and Hall/CRC, 2003.

Dirk Hovy, Taylor Berg-Kirkpatrick, Ashish Vaswani, and Eduard Hovy. Learning whom to trust with MACE. In *Proceedings of NAACL-HLT*, pages 1120–1130, 2013.

Yingyao Hu and Susanne M Schennach. Instrumental variable treatment of nonclassical measurement error models. *Econometrica*, 76(1):195–216, 2008.

Kosuke Imai and Teppei Yamamoto. Causal mediation analysis using R. *Advances in Social Science Research Using R*, pages 129–154, 2010.

Chris J Kennedy, Geoff Bacon, Alexander Sahn, and Claudia von Vacano. Constructing interval variables via faceted Rasch measurement and multitask deep learning: A hate speech application. In *arXiv preprint arXiv:2009.10277*, 2020.

Edward H Kennedy. Towards optimal doubly robust estimation of heterogeneous causal effects. *Electronic Journal of Statistics*, 17(2):3008–3049, 2023.

Reagan Mozer. Using large language models for expert surveys: Improved estimation with misclassified data. *Political Analysis*, 2026.

Nicholas Pangakis, Samuel Wolken, and Neil Fasching. Automated annotation with generative AI requires validation. *arXiv preprint arXiv:2306.00176*, 2023.

Table 13: Real-data subgroup misclassification rates.

Subgroup	n	Misclass.	FPR	FNR
plat0_long	624	0.266	0.10	0.55
plat0_short	624	0.337	0.15	0.58
plat23_long	624	0.303	0.12	0.52
plat23_short	624	0.383	0.18	0.50

Rebecca J Passonneau and Bob Carpenter. The benefits of a model of annotation. In *Transactions of the Association for Computational Linguistics*, volume 2, pages 311–326, 2014.

Jiayi Sui, Mingzhang Yin, and Sujit K Ghosh. Prediction-powered conditional inference. *arXiv preprint*, 2026.

Petter Törnberg. ChatGPT-4 outperforms experts and crowd workers in annotating political twitter messages with zero-shot learning. *arXiv preprint arXiv:2304.06588*, 2024.

Stefan Wager and Susan Athey. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242, 2018.

Caleb Ziems, William Held, Omar Shaikh, Jiaao Chen, Zhehao Zhang, and Diyi Yang. Can large language models transform computational social science? *Computational Linguistics*, 50(1):237–291, 2024.

A Cross-Model Heterogeneity Detail

Table 14: Cross-model $\Delta\pi$ summary (TweetEval sentiment, $K = 4$ text-feature subgroups).

Model	Gen.	Accuracy	$\Delta\pi$	Worst subgroup
GPT-3.5-turbo	3.5	0.72	0.198	True_medium
GPT-4-turbo	4	0.81	0.149	True_medium
GPT-4.1	4.1	0.86	0.098	True_medium
GPT-4o	4o	0.88	0.080	False_medium
GPT-4o-mini	4o-mini	0.83	0.136	True_medium
GPT-5.4	5.4	0.90	0.114	True_medium

B Adaptive Expert Allocation

Table 15: Allocation strategy comparison ($K = 4$, extreme, $n_{\text{expert}} = 250$).

Strategy	RMSE	|Bias|	Cov.	vs Uniform
Uniform	0.89	0.12	0.92	—
Uncertainty only	0.82	0.11	0.92	+7.9%
CATE-weighted	0.75	0.11	0.93	+15.7%
CATE $\times$ error	0.71	0.11	0.93	+20.2%

C Robustness to Subgroup Noise

Table 16: Coverage across subgroup label noise levels.

K	p_{flip}	$\text{Cov.}_{\text{naive}}$	$\text{Cov.}_{\text{global}}$	Cov._{HB}
2	0.00	0.875	0.925	0.958
2	0.25	0.890	0.932	0.950
2	0.50	0.897	0.945	0.945
4	0.00	0.904	0.930	0.956
4	0.25	0.909	0.948	0.960
4	0.50	0.919	0.944	0.946